

COMP3011J MC Project (Final)

22207232 Qiyue Zhu

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1 Brief Overview of the Open-Source App: [Kandean](#)

With the original version of [Kandean](#), users can effortlessly create, post, and share their unique recipes, meanwhile, comment, likes, and save various recipes.

The coming version of this app will also provide a digital fridge that tracks available ingredients. The possible coming features are: timely reminders for items nearing expiration, setting automatic updates for the quantity of each item stored and receive shopping suggestions when ingredients run low.

While most of the popular recipe applications focus on creating and sharing recipes or diet planning, [Kandean](#) uniquely integrates these functionalities with a brand-new digital fridge feature. Having this, you can easily manage your diet and shopping choices.

2 AI-supported Image Recognition (The First 2 Images in [1](#))

Complete the scan item function by introducing the image-recognition technique to automatically add new item to the digital fridge. This function is implemented by introducing **Baidu image recognition interface**, and we will use the most appropriate result (highest score) as the image recognition result.

By clicking the “scan item” floating button and choose an image of the ingredient you want to add, you can add that ingredient automatically. This process can be viewed in the video.

This part is implement in *DigitalFridgeFragment*, *InterfaceBaidu*, *BaiduResultBean*, *GetDiscernResultResponse*, and each function can be viewed in the slides since I think slides are better to match the class with each function.

3 Add Shopping List Pages [1](#)

Add a shopping list page supports handwriting or scan item’s image to add new items.

The corresponding pages contains:

- **Main View Page:** I add a shopping list button on the digital fridge page, by clicking that, you can enter this page, which will show you every item in your shopping list. The page is in *fragment_shopping_list.xml* and main functions are in *ShoppingListFragment*.
- **Add New Item:** This page can be entered from the shopping list pages by clicking the handwriting button or the scan button, same as the digital fridge pages. This page is in *fragment_create_shopping_item.xml* and functions are in *CreateShoppingItemFragment*.
- **Delete Item:** You can delete an item by clicking the red trash icon on the right-hand-side of that item. This page is in the page of view page, and functions are implemented in *AllShoppingAdapter*.

4 Auto-create A New Item If Needed (The Last Image In [1](#))

A new item can be created automatically when you lack the ingredient in your digital fridge when using the recipes. This function is another link between the recipe and the newly-added functions. In the stage 2 implement, I modify the recipe to let them have a “USED” button in the rear of each ingredient, now if the ingredient in your digital fridge is not sufficient, the app will automatically create one in your shopping list by redirecting you to the create shopping list item page and auto-filling the form. This is implemented by adding some parts related to translate to shopping list pages under the “ingredient not sufficient” case in *IngredientAdapter*.

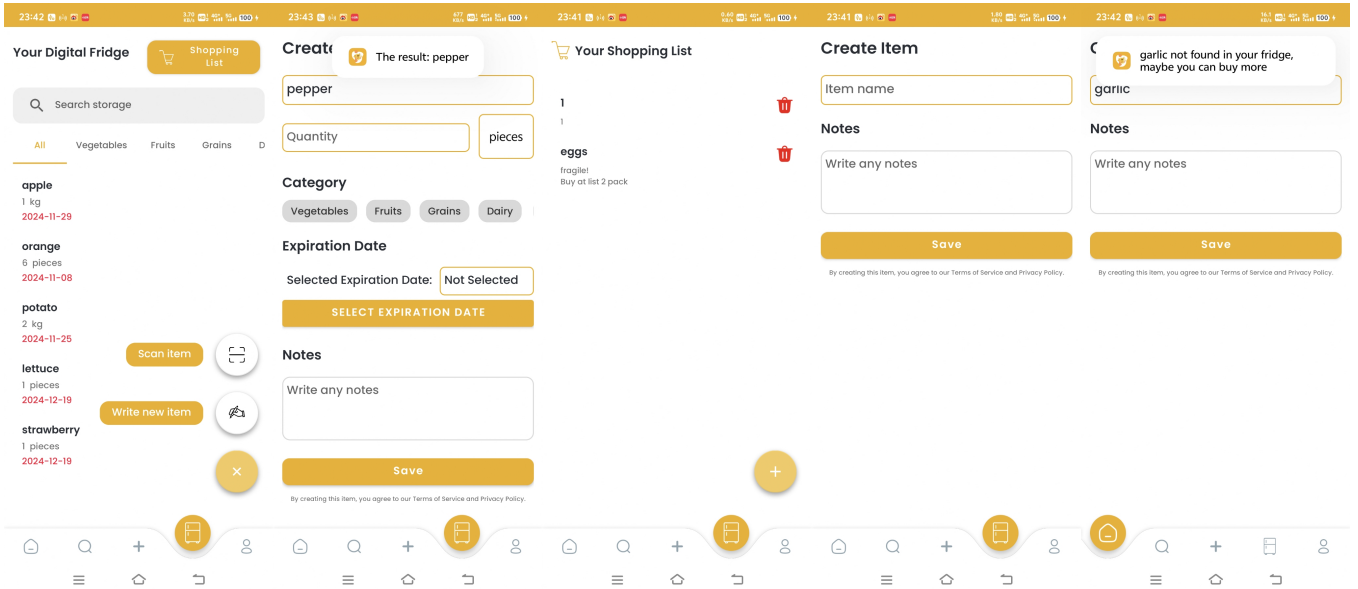


Figure 1: The pages of scan item and shopping list

5 Development Timeline 1

Time	Work to be completed	Comments
ALPHA SUBMISSION	Set up the new basic pages and layout of the digital fridge page on the main page	Basic new function setup based on the existing app
BETA SUBMISSION	Add the function of automatically decreasing the quantity of an item in the digital fridge if used in the recipe	Link the digital fridge with the recipe usage
FINAL SUBMISSION	Add an image-recognition API that enables users to scan items to add new items automatically, and add new shopping list pages which enable users to auto-create shopping list when lacking ingredients.	Add API and shopping list pages

Table 1: Timeline for implementation